

Nguyen Duong Hieu

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Education

Tran Phu High School for the Gifted, Natural Science 1 2020 – 2023

National Economics University, BS in Data Science in Economics and Business Sept 2023 – Now

- GPA: 3.81/4.0
- **Coursework:** Linear Algebra, Probability & Statistics, Optimization, Data Preparation & Storytelling, Data Structures & Algorithms, Machine Learning, Database Systems (SQL).

Experience

Research Assistant, DATAOPT Lab – National Economics University, Hanoi, Vietnam July 2024 – January 2026

- Investigated and implemented mathematical optimization algorithms (e.g., convex optimization, gradient-based methods) from research papers.
- Reproduced experiments to evaluate the theoretical properties and practical performance of numerical methods in machine learning contexts.
- Actively engaged in academic discussions to strengthen foundations in applied mathematics and data-driven decision making.

Research Assistant, Project: “Does Network Position Cause Scholarly Success?” Spring 2026
with Prof. Mike Nguyen (University of Southern California) and Prof. Minh Nguyen (Florida Atlantic University)

- Curated, cleaned, and structured large-scale academic collaboration datasets, resolving inconsistencies across multiple sources to ensure data validity.
- Collaborated with the research team to support empirical and network-based analysis by providing high-quality, analysis-ready datasets.

Teaching Assistant (Tutor), Python for Data Science & Power Bi, National Economics University – Hanoi, Vietnam August 2025 – December 2025

- Assisted the course instructor in preparing assignments and teaching materials for Python and Power BI courses.
- Evaluated and graded student submissions, providing detailed and constructive feedback to improve understanding.
- Supported students by explaining concepts in Python programming and data analysis, enhancing their practical skills in data science applications.

Publications

Promoting circular consumption behavior: Factors influencing Vietnamese Generation Z's intention to extend the lifespan of personal electronic devices 2026

Le Thu Thao, **Nguyen Duong Hieu**, Pham Phuong Trinh, Nguyen Minh Tung

22nd International Conference on Socio-economic and Environmental Issues in Development (ICSEED 2026)

- Investigated the determinants of Generation Z's intention to extend electronic device lifespans and the intention-behavior gap by integrating the Theory of Planned Behavior (TPB) and Push-Pull-Mooring (PPM) framework.
- Employed Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze survey data from 733 respondents.

Projects

Hanoi Temperature Forecasting System [GitHub] Late 2025

- Developed an end-to-end ML pipeline to forecast Hanoi's daily temperature (5-day horizon) using 10+ years of

weather data.

- Engineered 90+ features (lag variables, rolling statistics, STL decomposition, FFT) to capture temporal patterns and seasonality.
- Trained a multi-output CatBoost model using Optuna for hyperparameter tuning, achieving $R^2 \approx 0.84$.
- Deployed an interactive Streamlit application integrated with SHAP for model explainability and an MLOps pipeline for data drift detection (PSI, KS-test).
- *Technologies: Python, Pandas, Scikit-learn, CatBoost, Optuna, SHAP, ONNX, Streamlit*

Multimodal AI in Enterprise Robotic Process Automation

Spring 2026

- Designed a modular two-stage framework for multimodal information extraction and entity resolution to process unstructured business inputs.
- Integrated OCR and ASR (PhoWhisper) for data transcription, coupled with a zero-shot LLM (Gemini) for structured entity extraction.
- Developed a hybrid retrieval mechanism combining dense semantic representations (Sentence-BERT) with sparse lexical matching (BM25) via Reciprocal Rank Fusion (RRF).
- Achieved F1-scores of 0.91 (visual) and 0.85 (acoustic) under noisy conditions, maintaining a Recall@5 of 0.93 on a 50,000-SKU enterprise catalog.

Skills

Technical Skills

- Programming: Python (Pandas, NumPy, Matplotlib), SQL
- Machine Learning: CatBoost, model evaluation (RMSE, MAE, R^2), hyperparameter tuning (Optuna)
- Data Analysis: Exploratory Data Analysis (EDA), statistical analysis
- Mathematics: Optimization, Probability & Statistics, Linear Algebra
- Tools: Power BI, Streamlit, LaTeX
- Domain Knowledge: Economics (microeconomics, macroeconomics fundamentals)
- Research: Academic writing, paper reading, method reproduction

Certifications and Achievements

- **2025** Top 10, Hanoi Mathematical Modeling Competition
- **2024** Academic Encouragement Scholarship (Excellent), Faculty of Mathematical Economics, National Economics University
- **2024** Outstanding Youth Union Member Award, National Economics University
- **2023** Consolation Prize, Hai Phong City Mathematics Competition
- **2022** IELTS 7.0

Interests

- Machine Learning, Large Language Models (LLMs), and Multimodal AI
- Optimization Methods for Machine Learning
- Applied Mathematics for Data-Driven Decision Making